

Daily Report “Day 3”

Software	• Researched ROS nodes and WebSocket and built a prototype connecting ROS, WebSocket and React. • Simulated sensor data, detected unsafe conditions, and displayed live readings with connection status on the React interface. • Created and styled the main dashboard in GUI React. • Added live camera, manual/autonomous modes, the movement commands, motor speed and motion status indicator as well as sensor feedback sections. • Next: adding control commands and object detection models.
Firmware	• Wrote code of sensors: https://drive.google.com/drive/folders/1eBKijjUQG14MZxupfAb6fWBY2TPA7dBv?usp=drive_link • Bootloader and HC-05 communication were configured, and basic wireless transmission was successfully tested. Sensor firmware now reads voltage and current and sends combined data every 500 MS; next step is integrating the MPU-6050 IMU via I²C.
Hardware	• Completed the MCU board placement, layout, and design prototype with all required components and verified footprints based on manufacturer specifications. • Applied design rules: 0.8 mm clearance, 0.7 mm minimum trace width, and up to 1.4 mm for power/GND traces, with decoupling capacitors placed close to IC pins. • Verified the final PCB with 0 DRC errors and prepared it for toner-transfer fabrication.